Study of self-organization of ferrofluidic-polymer thin film in presence of an external magnetic/electric field

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PROPOSAL DETAILS

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Technical Details :

Scheme :	Core Research Grant		
Research Area :	Chemical Engineering (Engineering Sciences)		
Duration :	42 Months	**Contact No :**	+919717701531
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Nationality :	INDIAN	**Total Cost (INR)**	34,04,700

Is PI from National Laboratory/Research No

Project Summary :

Ferrofluids (FFs) are colloidal solutions, prepared with the initial motive to convert heat to work. Due to its response to an external magnetic field, it has applications in various diverse fields like aerospace, medicine, heat transfer, analytical instrumentation, etc. One of the important responses of FF to the magnetic field is Rosensweig instability, in which thicker film surfaces spontaneously deforms to form high asperity ratio structures. Another interesting response is due to a vertical temperature gradient in the presence of a perpendicular magnetic field. The gradient causes a gradient in magnetization which further changes the distribution of body force leading to convection. Thermo-magnetic convection is gaining importance due to its ability to enhance heat transfer where buoyancy-driven convection is negligible. If the layer of FF has a free interface, convection further will enhance due to the Marangoni effect. Though a lot of studies have been reported related to normal-field and Rayleigh -Benard-Marangoni instabilities in thicker (~ centimetres) and thinner (~100 μm) films, studies of instabilities in thinner films (less than 100 nm) are still missing. Moreover, due to the high asperity ratio of patterns, lubrication approximation is also no longer valid, and a standard framework of solving NSs equations by taking such approximations fails. At such scales, intermolecular interactions between the free interface of a carrier liquid and liquid-substrate interface are also not negligible. In the present project, our 1st objective is to study normal-field instability in ferrofluid films whose thickness is less than 100 nm. Navier-Stokes equations here will be solved using the LBM and Maxwell equations will be solved analytically or by using the FVM. In nm-ranged films considering FF as a single-phase magnetizable fluid is questionable. So, in the 2nd objective, FF will be considered as a 2-phase system. Governing equations of fluid flow will be solved by using the LBM and the particle motion will be tracked by Ladd's method. The 3rd objective will be to study thermomagnetic convection in FF thin films using the thermal Lattice Boltzmann method. And the final objective will be to study the stability of FF thin film under an electric field whose strength changes linearly with time. Ordered patterns of FFs formed through self-organization in the nm scale having high aspect ratios will have huge technological applications in fabrications of actuators in MEMS/NEMS, thin-film solar cells, optical diffraction gratings etc. Moreover, since the magnetic field is an external field, it can be altered externally and a pulsating motion can be set in, utilizable in creating artificial organs like the heart. The convective flow generation can be utilized in microfluidic devices to generate turbulence and planar self-organization of magnetic nano-particles due to a perpendicular electric field can be harnessed to create high-end sensors.

Objectives :

• 1. In this study, the ferrofluid thin film will be confined between a solid substrate and a bounding fluid. The ferrofluid will be considered as a magnetizable liquid and magnetic field will be applied normal to the ferrofluid-bounding fluid interface. i. Linear stability analysis to predict length and timescale of patterns. ii. To study how van der Waals interactions and magnetic field affects the pattern formation. 2. Study of Normal-field instability in ferrofluid thin films considering ferrofluid as two-phase solution. i. To study whether length scales changes if ferrofluid thin film is considered as two- phase mixture and the thickness/concentration limit where both the predictions match. ii. To study how the particles/ particle assemblies under magnetic field effects the dewetting of carrier fluid. 3. Study of Rayleigh -Benard-Marangoni instability in ferrofluid thin film. i. To study how the van der Waals interactions, magnetic field and temperature gradient change magnetic particle distribution inside the films and affect a Marangoni convection which in turn will affect the heat transfer. 4. Study of stability of ferrofluid thin film under electric field with increasing field strength with time. i. To study how magnetic particles cast as a bilayer over a carrier fluid gets affected and self-assemble due to an induced magnetic field created by a change in electric field in a direction perpendicular to the planar interface.

Keywords :
Self-organization, ferrofluid, magnetic field, thin film, Lattice Boltzmann Method

Expected Output and Outcome of the proposal :

The topic of this project finds application in the field of ferrofluid; is of considerable interest to the bio-polymer/thermal management fraternity at large and also has importance in the creation of smart fluids which can respond to non-contact external stimulii.The pattern formation of ferrofluid, miniaturization of these patterns and freezing of the pattern in the polymeric matrix (such that the patterns can stay even on the removal of the magnetic field) will open a new dimension for application in NEMS/MEMS, in sensor applications etc. On the other hand the easy control of the flow due to the external magnetic feld, will help in creating pulsating motion for biomedical devices, in the fabrication of actuators and so on. Therefore the results obtained from this research may be disseminated in the following ways: 1. Publications in leading conferences in this area and reputed journals. 2. Dissertations of the students working in this group. 3. Future work in this area leading to development of devices. It is planned to publish a technical report every six months of the project. The technical reports also can be termed as deliverables; will report the progress of research in the previous six months. The results will also direct newer experiments in the direction of fabrication of sensors for cancer detection and possible killing of cancerous cells through hyperthermia.

Suitability of the proposed work in major national initiatives of the Government:

Innovate India

Collaboration Details for last 5 Years :

Planned Collaboration for the proposed work with any foreign scientist/ **Yes**

Project Title: Study of self-organization of ferrofluidic-polymer thin film in presence of an external magnetic/electric field

Principal investigator: Dr. Jayati Sarkar, Indian Institute of Technology Delhi

OTHER TECHNICAL DETAILS

1. Origin of Proposal

The stable colloidal suspension of ferromagnetic or ferrimagnetic in a liquid were discovered in 1960s at NASA research centre by Steven Papell.[1] The absence of gravity in outer space created a challenge for the scientists to fuel the rocket engine and of magnetic fluid was invented so that the fuel could be controlled by strong magnets. The wide varieties of magnetic fluids were developed under the leadership of Ron Rosensweig of AVCO Corporation for commercial use. With commercialization of magnetic fluid, the name ferrofluid came into being in 1968 and was coined by Ron Rosensweig and R. Moskowitz.[2] Ferrofluids are different from magnetorheological fluids which are micron sized particles dispersed in fluid and behave as a solid in strong magnetic field.[3] But in ferrofluids magnetic particles and liquid coexist as a homogenous phase and hence consist of both the properties of magnet and liquid such as saturation magnetization, viscosity, volatility, thermal degradation and dieclectric behaviour. Ferrofluids are generally synthesized by size reduction and chemical precipitation.[4] Various methods were developed over the time to synthesize stable magnetic fluids for various technological applications. Ferrofluids has ample application in multidisciplinary area. They had been greatly used in dynamic sealing [4], heat dissipationc[4], doping and damping[4] and as lubricants[4] help in its place by magnetic field and also as actuator with dynamic magnetic fields used in ferrofluid based pump, valves and tunable optical system. The ferrofluids are of great importance in biomedical field and is used mainly in magnetic drug targeting, hyperthermia, contrast enhancement of Magnetic Resonance imaging – MRI, magnetic separation of cells and many more.[4-5] With advancement in biomedical research, developing ferrofluid in biocompatible polymer has become the main focus. With development in technology along with realistic models for numerical simulations associated with hydrodynamic forces has led to very realistic predictions of ferrofluid applications. Another interesting phenomenon observed in ferrofluid under magnetic field is interfacial instabilities. The applied magnetic field modify the droplet formation of ferrofluid which is an example of interfacial phenomenon. With the advancement in the micro-fabrication techniques many novel applications such as nanoscale self-assembly, microfluidic pumps and magnetic cell sorting. The flow behaviour and mechanism of ferrofluid droplets coupling with magnetic field and flow field opens new area of technological application. The magnetic field controlled droplet formation is of great interest in the field droplet based microfluidics. [6]There is great potential of usage of magnetic particles with polymer for the application of sensor and actuator [7]. Thus the fundamental study of ferrofluid as thin film and in thin polymeric film is of great importance in the arena of technological applications.

2. Review of status of Research and Development in the proposed area

2.1 International Status

Normal-field instability or Rosensweig instability was first observed by Cowley and Rosensweig in which free surface of a thick ferrofluid deforms spontaneously if magnetic induction (B_o) exceeds the critical value (B_c) and orientated perpendicularly. At small value of $B_o - B_c$, hexagonal configuration [8] of ferrofluid surface is shown in fig. 1, and at higher values, surface morphology changes into square configuration [9,10]. In the thin ferrofluid films, (magnetic force is the only destabilizing force whereas in thicker films gravity also contributes to the instability) first surface

deforms into the hexagonal pattern but upon increasing the magnetic field strength, droplets formed by rupturing the film while a preserving the hexagonal arrangement [11]. In the case of thin film of micro-magnetic drops with the thickness in the $O(10^2 - 10^3)\,\mu m$ (present on the dry plate), four different types of instability patterns are observed depending upon the droplet diameter, Bond number, and the field increasing rate [12-13]. On the substrates prewetted by miscible fluid, the size of the daughter droplet is inversely proportional to the square of magnetization strength (*M*), and at particular *M*, the number of daughter droplets is proportional to the area of the original drop [14]. Morphological features related to both normal-field instability and labyrinthine instability are observed if ferrofluid drop immersed in the miscible, non-magnetic fluid [15-16]. Numerically, normal-field instability in thicker films was studied using minimization of free energy or finite element method (FEM) and good agreement with the experimental results are observed if the relation between the magnetization and magnetic field is given by the Langevin function [17-20]. Thin film flow of ferrofluid on the inclined plane was studied numerically by simplifying the Navier-Stokes equations using lubrication approximation [21-25]. In the ferrofluid thin films where van der Wall interactions (destabilizing force) are dominant, satellite drop formed with increasing magnetic field and magnetization of fluid is considered as a linear function of the magnetic field [26].



Fig. 1Rosensweig peaks when $B_o > B_c$ in a vessel with diameter 120 mm. The picture is taken from [28].

In the normal-field instability, the magnetization of ferrofluid depends only on the magnetic field (***H***) only, but in general, it also depends on the temperature (*T*) and density (ρ). Non-zero gradients of ***H,*** *T*, and ρ change distribution of body force within the film which leads to the convection. Convection due to temperature gradient in the presence of a magnetic field is gaining importance due to the ability to enhance the heat transfer where buoyancy-driven convection is negligible. Finalyson et al. [27] theoretically studied the thermo-magnetic convection in the ferrofluid film confined between the two horizontal plates and the bottom plate was maintained at high temperature than the top plate. Critical Rayleigh number from this study is in agreement with the experimental work later carried out by Schwab et al [28]. Many theoretical and experimental studies are reported based on the different geometries (rectangular enclosure [29,30], cylinder [31], cube [32], two vertical plates [33]), different velocity and temperature boundary conditions [34] etc. In the case of free surface ferrofluids convection increases due to Marangoni effect. Linear and non-linear stability of combined buoyancy–surface tension effects in a ferrofluid layer heated from below has been analyzed by Qin et al. [35]. Weilepp et al [36] studied the coupling between Marangoni and Rosensweig instabilities in bilayer of ferrofluid and Newtonian liquid. Hennenberg et al. [37] studied the Rayleigh -Benard-Marangoni instability in the ferrofluid layer under vertical magnetic field.

Magnetic particles in the ferrofluid start interact under an external magnetic field if particles have a strong magnetic dipole moment (at higher concentration of particles or large size of particles). These interactions cause the formation of different patterns, chains, and aggregates leading to the enhanced thermal and rheological properties of the carrier liquid [38,39]. Observed formation of patterns in the ferrofluids under an external electric field if carrier liquid is a dielectric. Magnetic nanoparticles will have high permittivity and polarizability when compared to the insulating liquids. Due to the difference in charge density on either side of the nanoparticle results in induced dipole across it causing alignment with the applied electric field. From the electrostatic theory, induced electric dipoles in closed particles, from long chains under external field if other forces (van der Waals interactions, gravity, etc.) are not strong. These structures in the ferrofluid increase the dielectric breakdown voltage of carrier liquid [40-45].

Non-magnetic particle-free viscous thin films with a thickness less than 100 nm spontaneously deform due to intermolecular interactions and form periodic structures with the wavelength (λ_m) proportional to the square of the mean film thickness h_o. And also, a non-magnetic thin liquid film with the free interface deforms under perpendicular electric field and $\lambda_m \propto h_o^{3/2}$. Observed further reduction in the λ_m under electric field, if non-magnetic nanoparticles ($c_p < 1\%$) added to the film due to the increase in the dielectric constant of base film [46-48].

2.2 National Status

Shivakumar et al [49-50] studied the effect of different initial temperature gradients on the Marangoni ferroconvection in thicker ferrofilms to understand the parameters controlling the convection. Nanjundappa et al [51-53] studied the effect of magnetic field dependent viscosity and internal heat generation on Bénard–Marangoni convection in a horizontal ferrofluid layer.
In non-magnetic, particle free thin liquid films observed reduction in the length scale of patterns if viscosity (decreases with decrease in the thickness) of film depends on the thickness or moderate and strong slip conditions under van der Wall interactions [54-55]. On addition of nano particles to the film observed suppression of dewetting with the increase in concentration of particles due to the accumulation of particles at the substrate or at the free interface of film [56-62].

2.3 Importance of the proposed project in the context of current status

The literature gives us a fair idea of the extensive research on ferrofluid both theoretically and experimentally from the time of its invention by Steven Papell in 1965 at NASA for directing rocket fuel in outer space. The ample applicability of ferrofluid in all fields including biological applications gives a distinguished place to ferrofluid in the research field. The application of these magnetic fluids is based on the properties of absorbing electromagnetic energy at convenient frequency and its physical properties and direction changes with magnetic field. The ferrofluid develops fascinating patterns or self-assemble to form droplets induced by strong magnetic field and show no remanent magnetization once the magnetic field is removed. The proposed project initiates a fundamental study of ferrofluid as thin film and its potential applications in various fields.

The work so far has been limited to thicker films, where gravitational force plays an important role, and the underlying fluid used was a colloidal one. In order to understand how the fluid will really behave in microgravity situation, for which ferrofluid was originally invented, one needs to delve into thinner film dynamics where gravity does not play an important role and forces like van der Waals, surface tension come into play. Ferrofluid as a thin film has not been exploited to large extent. Past numerical studies has been carried out using free energy minimization principle or finite element method. While free energy minimization principle doesn't give detailed dynamics of instabilities, solving Navier-Stokes (NS) equations by finite element method on the other hand is computationally highly expensive. Moreover as the experiments suggest even for thicker films the asperity ratio of the patterns formed are quite high, so a long wave approximation will surely fail in such circumstances. Thus, though the need of an instability study in thinner ferrofluidic film (which is anticipated to generate high asperity-ratio patterns as very snall lengthscales) is well understood, a means to do so has not been deviced so far, as evident from the negligible amount of study found in literature.

The inception of the idea of ferrofluid as a single thin film and also as a suspension of magnetic particles within a liquid carrier film will be investigated using thin film dynamics and Lattice Boltzman method and that will be the focal point of study in this proposed project. The fundamental approach of thin film dynamics on ferrofluid will generate different morphologies and smaller length-scale patterns of ferrofluid. These smaller patterns can be greatly used in microfluidics and

also in NEMS/MEMS. Miniaturization of these patterns will open ample opportunities in the above mentioned branch of applicability. The thermally induced magnetic flow of these magnetic particles in a liquid might create turbulence in the microfluidic channels, which is a much soughted phenomena at such lengthscales. The non-contact heating of the ferrofluid may create thermal heat engines in NEMS/MEMS and also may increase the thermomagneto mobility of the magnetic particles in a liquid. The pulsating flow of the ferrofluid that increases the convective heat transfer can be controlled by applying heat over thin polymeric layer which has the potential to induce enhanced pulsating flow in artificial organs. Moreover in this proposed research the magnetic nanoparticles are incorporated in thin polymeric film and the self-organization of the thin polymeric film can induce self-assembly of the magnetic nanoparticles, which can arrange themselves in a particular pattern with/without the aid of the magnetic field and will be studied for the first time. The self-assembly of the magnetic nanoparticles without being induced by the permanent magnet will give added advantage to these arrays of magnetic particles for various applications in magneto-optic sensors, temperature sensor or as actuators. These structures encapsulated within biocompatible polymer can be used for various biomedical applications.

3. Work Plan

3.1 Methodology

i. Study of Normal-field instability in ferrofluid thin films considering ferrofluid as single-phase magnetizable liquid

Considered ferrofluid thin film (mean film thickness $(h_o) < 100\,nm$) bounded by solid substrate and fluid, schematic representation of system is shown in fig.2.

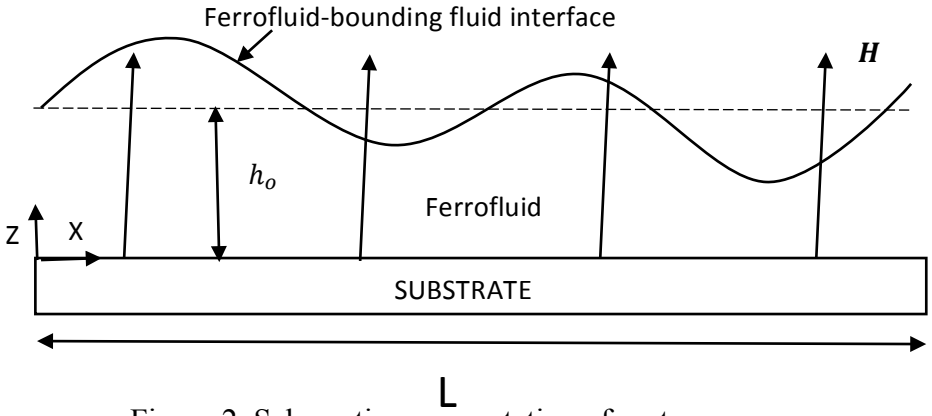


Figure 2. Schematic representation of system

Uniform magnetic field applied perpendicular to the film. Film is laterally unbound, non-draining with constant interfacial tension. For incompressible flow, equations of motion are given by

$$\nabla . \mathbf{v} = 0, \tag{1}$$

$$\rho \frac{\mathrm{D}\mathbf{v}}{Dt} = -\nabla p + \eta \nabla^2 \mathbf{v} + \nabla \cdot \boldsymbol{T_m} + F_e. \tag{2}$$

p, F_e, $\boldsymbol{T_m}$ are thermodynamic pressure, external forces other than magnetic force, and Maxwell stress respectively.

Here viscous stress tensor is considered as symmetric this is valid, if magnetization **M** and magnetic fields **H** are colinear. Collinearity assumption is true for particles of sufficiently small size so that magnetization direction can rotate freely with in the particle. For large size of particles magnetic moment is locked to the orientation of the particle and if direction of **H** changes, **M** responds by rotating the particle, which is resisted by the fluid viscous-drag torque. In that case body couple $\mu_o \mathbf{M} \times \boldsymbol{H}$ will have some finite value and results in asymmetric viscous stress. And ferrofluid is considered as magnetizable, non-conducting liquid, for this case Maxwell stress is given by following equation.

$$\boldsymbol{T_m} = -\left\{\int_{\mathbf{0}}^{\boldsymbol{H}} \mu_o \left[\frac{\partial(vM)}{\partial v}\right]_{H,T} dH + \frac{1}{2}\mu_o H^2\right\}\boldsymbol{I} + \boldsymbol{BH}. \tag{3}$$

$\boldsymbol{B}, \mu_o, v$ are magnetic induction, permeability of vacuum and specific volume.

Solid substrate and bounding fluid are non-magnetizable with equal magnetic permeabilities and there are no charges in the bounding fluid. Under these assumptions, $\boldsymbol{B_j}$ (*j=1* for ferrofluid film, 2 for bounding fluid and 3 for solid substrate) $\boldsymbol{H_j}$ are governed by steady Maxwell equations.

Gauss Law for magnetism

$$\boldsymbol{\nabla} \cdot \boldsymbol{B_j} = 0. \tag{4}$$

Ampere's Law if there are no currents

$$\boldsymbol{\nabla} \times \boldsymbol{H_j} = 0. \tag{5}$$

Relation between $\boldsymbol{B_j}$, $\boldsymbol{H_j}$, and $\boldsymbol{M_j}$ changes depending upon the phase.

$$\boldsymbol{B_j} = \begin{cases} \mu_o\left(\boldsymbol{H_j} + \boldsymbol{M_j}\right) \text{ for } j = 1 \\ \mu_o \boldsymbol{H_j} \text{ for } j = 2,3 \end{cases} \tag{6}$$

Magnetization of magnetic liquid depends on temperature, density and magnetic field. But here $\boldsymbol{M}$ is function of $\boldsymbol{H}$ only.

$$\boldsymbol{M_1} = \begin{cases} \chi \boldsymbol{H_1} & \text{if ferrofluid is linearly magnetizable} \\ M_s\left[\coth(\xi H) - \frac{1}{\xi H}\right]\frac{\boldsymbol{H_1}}{H_1} & \text{if ferrofluid is non} - \text{linearly magnetizable} \end{cases} \tag{7}$$

Susceptibility is defined as $\chi = \frac{\boldsymbol{M}}{\boldsymbol{H}}$ and $\xi = \frac{3\chi_o}{M_s}$. M_s is saturation magnetization and χ_o is initial Susceptibility. From eq. (5), magnetic field is irrotational, so there exists a magnetic scalar potential ψ which is defined by

$$\boldsymbol{H_j} = -\boldsymbol{\nabla}\psi_j. \tag{8}$$

By taking dot product of eq. (8)

$$\nabla^2\psi_j = \begin{cases} 0, & \text{if } \mathbf{M_j} \text{ is linear function of } \mathbf{H} \text{ or zero} \\ \boldsymbol{\nabla} \cdot \boldsymbol{M_1}, & \text{otherwise} \end{cases} . \tag{9}$$

Magnetic field boundary conditions:

From eq. (4) normal component of $\boldsymbol{B}$ and from eq. (5) tangential component of $\boldsymbol{H}$ are continuous at the interface.

$$\boldsymbol{n} \cdot \left(\boldsymbol{B_1} - \boldsymbol{B_{j'}}\right) = 0, \; j' = 2 \,\&\, 3 \tag{10}$$

$$\boldsymbol{n}\times\left(\boldsymbol{H}_{j'}-\boldsymbol{H}_{1}\right)=0 \tag{11}$$

Where $\boldsymbol{n}$ is vector normal to the interface. And also, tangential component of Maxwell stress due to magnetic field is continuous at the interface.

$$\left(\boldsymbol{n}\cdot\boldsymbol{T}_{\boldsymbol{m}}\cdot\boldsymbol{t}\right)_{1}=\left(\boldsymbol{n}\cdot\boldsymbol{T}_{\boldsymbol{m}}\cdot\boldsymbol{t}\right)_{j'} \tag{12}$$

In this project, equations of motion are solved by Lattice Boltzmann method based on Pseudopotential model and Maxwell equations are solved either by analytically or Finite volume method.

Lattice Boltzmann method (LBM)

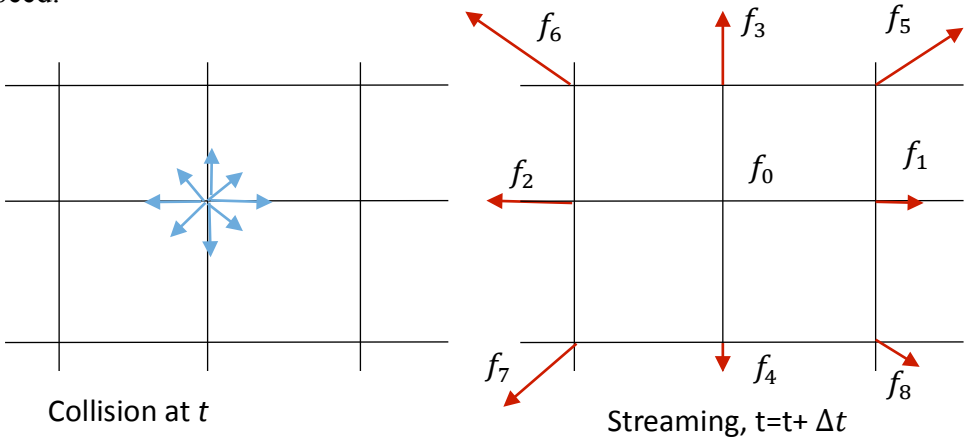
Discretized velocity Boltzmann equation with BGK collision operator when external force is zero is given by

$$f_i\left(\vec{x}+\vec{c_i}\Delta t,\ t+\Delta t\right)-f_i\left(\vec{x},\ t\right)=-\frac{\Delta t}{\tau}\left(f_i\left(\vec{x},\ t\right)-f_i^{eq}\left(\vec{x},\ t\right)\right)+S_i\Delta t \tag{13}$$

$$f_i^{eq}\left(\vec{x},\ t\right)=\ w_i\rho\left(1+\frac{\vec{c_i}.\ \vec{u}}{c_s^2}+\frac{\left(\vec{c_i}.\ \vec{u}\right)^2}{2c_s^4}-\frac{\vec{u}.\ \vec{u}}{2c_s^2}\right) \tag{14}$$

$$\tau=\left(\frac{\upsilon}{c_s^2\Delta t}+\frac{1}{2}\right)\Delta t$$

$f_i\left(\vec{x},\ t\right)$ is density of particles at position $\vec{x}$, with lattice speed $\vec{c_i}$ at particular t and υ is kinematic viscosity. w_i depends on velocity model, for D2Q9 model $w_0=\frac{4}{9}, w_1{=}w_2{=}w_3{=}w_4=\frac{1}{9}$, $w_5{=}w_6{=}w_7{=}w_8=\frac{1}{9}$. S_i is total force dependent on lattice speed.



Schematic representation of D2Q9 velocity model

Macroscopic variables are calculated by

$$\rho=\sum_i f_i \tag{15}$$

$$\rho\vec{u}=\sum_i \vec{c}_i f_i+\ \frac{\vec{F}\Delta t}{2} \tag{16}$$

External force incorporated by scheme proposed by Guo et al., 2002.

$$S_i=\left(1-\frac{\Delta t}{2\tau}\right)w_i\left[\frac{\vec{c}_i-\vec{u}}{c_s^2}+\frac{(\vec{c}_i.\vec{u})\vec{c}_i}{c_s^4}\right].\ \vec{F} \tag{17}$$

Pseudopotential method was proposed by as Shan and Chen in 1993 based on the fact that liquid and vapor will form depending on the interaction strength between them.

$$\boldsymbol{F_{f-f}}(\vec{x},t) = -G\psi(\vec{x},t)\sum_i w(|c_i|^2)\psi(\vec{x}+\vec{c_i}\delta_t,t)\vec{c_i} \quad (18)$$

G is interaction strength, if it is less than zero force between two particles is attractive and $\psi(\vec{x},t)$ is interaction potential. Interaction force between fluid and substrate is given by

$$\boldsymbol{F_{f-s}}(\vec{x},t) = -G_{ads}\psi(\vec{x},t)\sum_i w_i S'\left(\vec{x}+\vec{c_i}\Delta t,\ t\right)\vec{c_i}\ . \quad (19)$$

$S'\left(\vec{x}+\vec{c_i}\Delta t,\ t\right)$ is equal to one if $\vec{x}+\vec{c_i}\Delta t$ is a solid site, 0 otherwise. Force due to van der Wall interaction with in the ferrofluid film is given by

$$\vec{F}_{vdW}(\vec{x},t) = -\nabla\emptyset = \sum_i w(|c_i|^2)\emptyset(\vec{x}+\vec{c_i}\delta_t,t)\vec{c_i}\ . \quad (20)$$

$$\phi(h) = \frac{A_{eff}}{6\pi h^3}\left(1-\left(\frac{l_o}{h}\right)^6\right) \quad (21)$$

Magnetic force is given by

$$\boldsymbol{F_{m,j}} = -\boldsymbol{\nabla}\left\{\int_{\boldsymbol{0}}^{\boldsymbol{H}} \mu_o\left[\frac{\partial(v\boldsymbol{M_j})}{\partial v}\right]_{H,T} dH\right\}\boldsymbol{I} + \mu_o\boldsymbol{M_j}\boldsymbol{\nabla}. \quad (22)$$

Net force, $\boldsymbol{F} = \boldsymbol{F_{f-f}} + \boldsymbol{F_{f-s}} + \boldsymbol{F_{vdw}} + \boldsymbol{F_{m,j}}$.

Boundary conditions:

Periodic boundary and half-way boundary conditions are applied in lateral direction, vertical direction respectively.

ii. Study of Normal-field instability in ferrofluid thin films considering ferrofluid as two-phase solution

If the film thickness is in the order of nano-meter range considering ferrofluid as a single-phase magnetizable fluid is questionable. So, in the second objective ferrofluid is considered as a two-phase system. This problem will be solved by two methods.

Method I:

Carrier liquid, bounding fluid and magnetic particles are considered as three different components.

$$f_i^k\left(\vec{x}+\vec{c_i}\Delta t,\ t+\Delta t\right) - f_i^k\left(\vec{x},\ t\right) = -\frac{\Delta t}{\tau}\left(f_i^k\left(\vec{x},\ t\right) - f_i^{k,eq}\left(\vec{x},\ t\right)\right) + S_i^k\Delta t \quad (23)$$

$k = 1$ for carrier liquid, 2 for magnetic particles and 3 for bounding phase. Macroscopic variables are calculated depending upon distribution of each component. G and G_{ads} will depend on the component and van der Wall force is present in the carrier fluid (because of form of disjoining pressure in the eq. (21)). Magnetic force is given by

$$\boldsymbol{F_m^k} = -\rho^k(x)\sum_{i=0}^{8} G_a^k\,\rho^k(x+\boldsymbol{c_i}dt)\boldsymbol{c_i}dt\ , \quad (24)$$

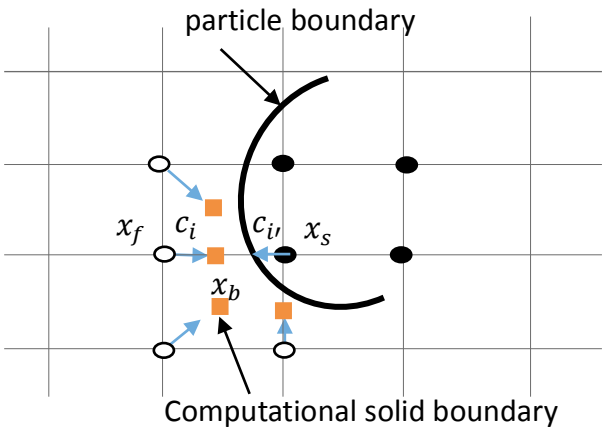
$$G_a^k = \begin{cases} w_a\left[\frac{m^2}{|c_i|^3} - \frac{(\boldsymbol{m}\cdot\boldsymbol{c_i})^2}{|c_i|^5}\right], & for\ k = 2. \\ 0, for\ k = 1,3 \end{cases} \quad (25)$$

Where $\boldsymbol{m}$ is magnetic moment of particle and it is given by $\boldsymbol{m} = \mu_o\boldsymbol{M}a_d d$ and $a_d d$ is volume of the particle.

Method II:

Governing equations of fluid flow are solved by using the Lattice Boltzmann method (eq. 13-eq. 21) but in the fluid phase force due to magnetic field is zero and each particle is tracked by

Ladd's method. In this method fluid is assumed to be present the inside particle also and at the at the fluid-particle (f-p) boundary half-way bounce back condition is used. Reflected distributions at the f-p boundary is given



$$f_{i'}(x_f, t+\Delta t) = f_i^*(x_f, t) + 2w_i \frac{\rho}{c_s^2}(\vec{c}_{i'} \cdot \vec{u}_b) \quad (26)$$

$$\vec{u}_b = \vec{u}_p + \vec{\omega}\times(\vec{x}_b - \vec{x}) \quad (27)$$

Where $\vec{x}$ is center of mass of the particle.

$$\overrightarrow{F'}\left(x_b, t+\frac{\Delta t}{2}\right) = 2\frac{\delta_x^D}{\delta t}\left[f_i^*(x_f, t) - f_i^*(x_s, t) + 2w_i \frac{\rho}{c_s^2}(\vec{c}_i \cdot \vec{u}_b)\right] \vec{c}_i \quad (28)$$

$$\vec{F}_{p,drag}\left(t+\frac{\Delta t}{2}\right) = \sum_{x_b} \overrightarrow{F'}\left(x_b, t+\frac{\Delta t}{2}\right) \quad (29)$$

$$\vec{F}_{p,drag}(t+\Delta t) = 0.5\left(\vec{F}_p\left(t-\frac{\Delta t}{2}\right) + \vec{F}_p\left(t+\frac{\Delta t}{2}\right)\right)$$

Magnetic force on the particle is $\boldsymbol{F_{p,mag}} = \boldsymbol{\nabla(m.H)}$ (30)

To suspend magnetic particles in the carrier liquid, they are coated with the surfactant layer. So, repelling force between two particles is

$$\boldsymbol{F_{p,r}} = \frac{3\mu_o \mathrm{m}^2}{2\pi(2\mathrm{R})^4} e^{-\left[\beta\left(\frac{r}{2R}-1\right)\right]\hat{r}} \quad (31)$$

Rate of repelling force is controlled by β and r is relative position between the two vectors.

$$\boldsymbol{F_{p,tot}} = \boldsymbol{F_{p,drag}} + \boldsymbol{F_{p,mag}} + \boldsymbol{F_{p,r}} \quad (32)$$

$$\boldsymbol{T_p}(t) = \sum_{x_b}(\vec{x}_b - \vec{x})\times\boldsymbol{F_{p,tot}}\left(x_b, t+\frac{\Delta t}{2}\right) \quad (33)$$

Velocity and position of each particle is updated by Newton's law of motion.

$$\boldsymbol{F_{p,tot}} = M_p \frac{d\boldsymbol{u_p}}{dt} \quad (34)$$

$$\boldsymbol{T_p} = I_p \cdot \frac{d\boldsymbol{\omega_p}}{dt} \quad (35)$$

$$\frac{d\boldsymbol{x_p}}{dt} = u_p \quad (36)$$

iii. Study of Rayleigh-Bernard-Marangoni field instability in ferrofluid thin film

In this objective magnetic fluid is considered as magnetizable liquid and magnetization of liquid is a function of temperature and magnetic field and it is given by

$$\mathbf{M} = \frac{\boldsymbol{H}}{H}[M_o + \chi(H - H_o) - K(T - T_o)] \quad (37)$$

$K\left(-\left[\frac{\partial \boldsymbol{M}}{\partial T}\right]_{H_o}\right)$ is pyromagnetic coefficient, $\chi = \left[\frac{\partial \boldsymbol{M}}{\partial H}\right]_{T_a}$ and M_o is calculated at H_o, T_a. Energy equation is given by

$$\left[\rho_o C_{v,H} - \mu_o \boldsymbol{H} \cdot \left[\frac{\partial \boldsymbol{M}}{\partial T}\right]_{H,v}\right]\frac{DT}{Dt} + \mu_o T \left[\frac{\partial \boldsymbol{M}}{\partial T}\right]_{H,v} \cdot \frac{D\boldsymbol{H}}{Dt} = k_1 \nabla^2 T \tag{38}$$

Governing equations of motion and Maxwell equations are solved as the method described to solve the first objective. Energy equation is solved by

$$g_i\left(\vec{x} + \vec{c_i}\Delta t,\ t + \Delta t\right) - g_i\left(\vec{x},\ t\right) = -\frac{\Delta t}{\tau_T}\left(g_i\left(\vec{x},\ t\right) - g_i^{eq}\left(\vec{x},\ t\right)\right) + S_i \Delta t \tag{39}$$

$$\rho T = \sum_i g_i \tag{40}$$

$$g_i^{eq}\left(\vec{x},\ t\right) = w_i \rho T \left(1 + \frac{\vec{c_i}.\ \vec{u}}{c_s^2} + \frac{\left(\vec{c_i}.\ \vec{u}\right)^2}{2c_s^4} - \frac{\vec{u}.\ \vec{u}}{2c_s^2}\right) \tag{41}$$

Boussinesq approximation is used to calculate fluid properties (ρ, ϑ). Bottom surface is maintained at high temperature than the top surface.

iv. Study of stability of ferrofluid thin film under electric field with increasing electric field strength

Ferrofluid (1) and bounding fluids (2) are perfect dielectrics and sandwiched between two electrodes. In the bounding phase electric field increases with the time. Fluid flow will be solved by Pseudopotential LBM as described in the methodology objective I, Maxwell equations due to electric field will be solved either analytically or by finite volume method, and particles motion tracked by Ladd's method.

Maxwell equations due to electric field:

$$\boldsymbol{\nabla} \cdot \boldsymbol{E_j} = 0, \quad j = 1,2 \tag{42}$$

$$\boldsymbol{E_j} = -\boldsymbol{\nabla}\psi_j^{el} \tag{43}$$

By taking divergence of eq. (42)

$$\nabla^2 \psi_j^{el} = 0 \tag{44}$$

Maxwell stress due to electric field

$$\boldsymbol{T}_{el,j} = \epsilon_j \epsilon_o \left[\boldsymbol{E_j E_j} - \frac{1}{2}\left(\boldsymbol{E_j} \cdot \boldsymbol{E_j}\right)\boldsymbol{I}\right] \tag{45}$$

Force due to an external electric field

$$\boldsymbol{F_{el}} = \nabla \cdot \boldsymbol{T_m} \tag{46}$$

Interfacial conditions:

$$\psi_1^{el} = \psi_2^{el} \tag{47}$$

$$\epsilon_2\left(-\boldsymbol{\nabla}\psi_2^{el} \cdot \boldsymbol{n}\right) - \epsilon_1\left(-\boldsymbol{\nabla}\psi_1^{el} \cdot \boldsymbol{n}\right) = q \tag{48}$$

Dipole moment of particle

$$\boldsymbol{P} = \frac{1}{4\pi\epsilon_o}\left(\frac{\epsilon_p - \epsilon_1}{\epsilon_p + 2\epsilon_1}\right) R_p^3 \boldsymbol{E} \tag{49}$$

Interaction between the two particles located at $\boldsymbol{x}$ and $\boldsymbol{x'}$

$$U(\boldsymbol{x}, \boldsymbol{x'}) = -\frac{P^2}{2\pi\epsilon_1 |x - x\prime|^3} \tag{50}$$

3.2 Time Schedule of activities giving milestones through BAR diagram.

Objective 1

1(A) Linear stability analysis

1(B) Developing code using Lattice Boltzmann method and validation of code

1(C) Non-linear simulations and experiments

1(D) Analysis of results

1(E) Documentation of results

Objective 2

2(A) Developing code using Ladd's method to track particles and validation of code

2(B) Non-linear simulations of magnetic particles laden thin film

2(C) Analysis of results

2(D) Documentation of results

Objective 3

3(A) Developing code using thermal Lattice Boltzmann method to account for temperature effects and validation of code

3(B) Non-linear simulations and experiments

3(C) Analysis of results

3(D) Documentation of results

Objective 4

4(A) Developing code using Lattice Boltzmann method to account for electric effects and validation of code

4(B) Non-linear simulations and experiments

4(C) Analysis of results

4(D) Documentation of results

Time schedule of activities

Activity	Mon 1-6	Mon 7-12	Mon 13-18	Mon 19-24	Mon 25-30	Mon 31-36
1(A)	✓					
1(B)	✓					
1(C)	✓					
1(D)	✓					

1(E)	✓					
2(A)		✓				
2(B)		✓				
2(C)		✓	✓			
2(D)		✓	✓			
3(A)			✓	✓		
3(B)			✓	✓		
3(C)			✓	✓	✓	
3(D)			✓	✓	✓	
4(A)					✓	✓
4(B)					✓	✓
4(C)					✓	✓
4(D)					✓	✓

3.3 Suggested plan of action for utilization of research outcome expected from the project

The topic of this project finds application in the field of ferrofluid; is of considerable interest to the bio-polymer/thermal management fraternity at large and also has importance in the creation of smart fluids which can respond to non-contact external stimulii.The pattern formation of ferrofluid, miniaturization of these patterns and freezing of the pattern in the polymeric matrix (such that the patterns can stay even on the removal of the magnetic field) will open a new dimension for application in NEMS/MEMS, in sensor applications etc. On the other hand the easy control of the flow due to the external magnetic feld, will help in creating pulsating motion for biomedical devices, in the fabrication of actuators and so on. Therefore the results obtained from this research may be disseminated in the following ways:

1. Publications in leading conferences in this area and reputed journals.
2. Dissertations of the students working in this group.
3. Future work in this area leading to development of devices.

It is planned to publish a technical report every six months of the project. The technical reports also can be termed as deliverables; will report the progress of research in the previous six months. The results will also direct newer experiments in the direction of fabrication of sensors for cancer detection and possible killing of cancerous cells through hyperthermia.

3.4 Environmental impact assessment and risk analysis

Ferrofliuid, the colloidal suspension of nanoscale magnetic particles in a liquid is highly suitable for various applications. Various methods are now developed to create ferrofluid which are environment friendly and these particles are either water based or oil based. The ferrofluids can be made

biocompatible by using non-toxic bio friendly coating. Ferrofluid are greatly used for biomedical purposes and for targeted drug delivery as they are mostly non-toxic and environmental friendly. Ferrofluid steered under magnet is also developed to give rise to new tools for environmental engineering and safety. These nanoparticles are considered generally biocompatible as they are used in biological assays and cell monitoring. Moreover ferrrofluid has been used for decays to separate valuable mineral ores by magneto-gravimetric method which implies that ferrofluid is environmental friendly and has no risk to the natural surroundings. Thus the magnetic nanoparticles are biologically compatible, bio-friendly and have no reverse impact on the surrounding environment and are non-toxic. Thus ferrofluid is one of the most risk free nanoparticle to work under the laboratory safety measures.

2.1 Expertise available with the investigators in executing the project

The principal investigator's research interests lie in the field of interfacial science, smart materials, self-assembly, rheology and complex fluids. She has been involved with the research of polymeric films, adhesion-debonding, dewetting of especially thin films from her PhD days in Indian Institute of Technology, Kanpur. She has worked both in the numerical and experimental domain. In recent years, she is working on self-assembly of graphene nano-particles at the polymer interface.

4.2 Summary of roles/responsibilities for all Investigators

S. No.	Name of the Investigators	Roles/Responsibilities
1.	Dr. Jayati Sarkar	Principal Investigator will be solely responsible for the execution of the project

4.3 Key publications published by the investigators pertaining to the theme of the proposal during the last 5 years.

Sl. No.	*Title of Paper(s)*	*Name of the Co-Author(s)*	*Name of the Journal*	*Page(s)*	*Volume & Year*
1	Miniaturization of surface patterns in soft elastic film over patterned substrates	S. Basu, J Sarkar	Chemical Engineering Science	195-203	197, 2019
2	Hierarchical micro-and nanofabrication by pattern-directed contact instabilities of thin viscoelastic films	A Ghosh, D Bandyopadhyay, J Sarkar, A Sharma	Physical Review Fluids	124004	2017

3	**A Finite Element Study of Adhesion of Soft Thin Elastic Films cast on Rough Surfaces**	**S. Mishra,H.Annepu and J. Sarkar**	**International Journal of Adhesion and Adhesives**	**102–110**	**79, 2017**
4	**Kinetics of sub-spinodal dewetting of thin films of thickness dependent viscosity**	**T. Rao. Kotni, R. Khanna nad Jayati Sarkar**	**Journal of Physics: Condensed Matter**	**175001**	**29, 2017**
5	**Mechanical Strain Induced Tunable Anisotropic Wetting on Buckled PDMS Silver Nanorods Arrays.**	**P. Goel , S. Kumar , J. Sarkar , and J. P. Singh**	**ACS APPL. MATER. INTERFACES,**	**8419–8426**	**7 (16), 2015**
6	**Miniaturized Pattern Formation in Elastic Films Cast on Sinusoidally Patterned Substrates.**	**H. Annepu**	**LANGMUIR**	**12278-12286**	**30 (41), 2014**
7	**Kinetically engendered subspinodal length scales in spontaneous dewetting of thin liquid films**	**T. Rao Kotni, And R. Khanna.**	*PHYSICAL REVIEW E, RAPID COMMUNICATIONS,* **Impact Factor:** *2.326* **Number of Citations:2** **The standards for Rapid Communication are parallel those for PRL but is of interest to a more specialized readership.**	**020401 (R)**	**90, 2014**
8	**Pattern Formation in Soft Elastic Films Cast on Periodically Corrugated Surfaces Linear Stability and Finite Element Analysis**	**H. Annepu , And S. Basu.**	*Modelling Simul. Mater. Sci. Eng.*	**055003**	*22(5), 2014*

4.4 Bibliography

1.SPINOFF, I. The Ferrofluids Story.

2. https://www.magdev.co.uk/ferrofluid

3.Olabi, A. G., & Grunwald, A. (2007). Design and application of magneto-rheological fluid. *Materials & design*, *28*(10), 2658-2664.

4. Scherer, C., & Figueiredo Neto, A. M. (2005). Ferrofluids: properties and applications. *Brazilian Journal of Physics*, *35*(3A), 718-727.
5. Shokrollahi, H. (2013). Structure, synthetic methods, magnetic properties and biomedical applications of ferrofluids. *Materials Science and Engineering: C*, *33*(5), 2476-2487.
6. Shi, D., Bi, Q., & Zhou, R. (2014). Numerical simulation of a falling ferrofluid droplet in a uniform magnetic field by the VOSET method. *Numerical Heat Transfer, Part A: Applications*, *66*(2), 144-164.
7. Ding, Z., Wei, P., Chitnis, G., & Ziaie, B. (2011). Ferrofluid-impregnated paper actuators. *Journal of microelectromechanical systems*, *20*(1), 59-64.
8. Cowley, M. D. & Rosensweig, R. E. 1967 The interfacial stability of a ferromagnetic fluid. *J. Fluid Mech.* 30, 671.
9. Gailitis, A. 1977 Formation of the hexagonal pattern on the surface of a ferromagnetic fluid in an applied magnetic field. *J. Fluid Mech.* 82 (3), 401–413.
10. Abou, B., Wesfreid, J. E., & Roux, S. (2000). The normal field instability in ferrofluids: hexagon–square transition mechanism and wavenumber selection. *Journal of Fluid Mechanics*, *416*, 217-237.
11. Berkovsky, B., & Bashtovoi, V. (1980). Instabilities of magnetic fluids leading to a rupture of continuity. *IEEE Transactions on Magnetics*, *16*(2), 288-297.
12. Chen, C. Y., & Lo, L. W. (2006). Breakup of thin films of micro-magnetic drops in perpendicular fields. *Journal of magnetism and magnetic materials*, *305*(2), 440-447.
13. Chen, C. Y., & Lo, L. W. (2006). Influences of field conditions on the rup turing instability of a circular thin magnetic film. *Magnetohydrodynamics*, *42*(1), 31-40.
14. Chen, C. Y., & Cheng, Z. Y. (2008). An experimental study on Rosensweig instability of a ferrofluid droplet. *Physics of Fluids*, *20*(5), 054105.
15. Chen, C. Y., & Li, C. S. (2010). Ordered microdroplet formations of thin ferrofluid layer breakups. *Physics of fluids*, *22*(1), 014105.
16. Chen, C. Y., Tsai, W. K., & Miranda, J. A. (2008). Hybrid ferrohydrodynamic instability: coexisting peak and labyrinthine patterns. *Physical Review E*, *77*(5), 056306.
17. Chen, C. Y., Yang, Y. S., & Miranda, J. A. (2009). Miscible ferrofluid patterns in a radial magnetic field. *Physical Review E*, *80*(1), 016314.
18. Matthies, G., & Tobiska, L. (2005). Numerical simulation of normal-field instability in the static and dynamic case. *Journal of Magnetism and Magnetic Materials*, *289*, 346-349.
19. Gollwitzer, C., Matthies, G., Richter, R., Rehberg, I., & Tobiska, L. (2007). The surface topography of a magnetic fluid: a quantitative comparison between experiment and numerical simulation. *Journal of fluid mechanics*, *571*, 455-474.
20. Lange, A., Richter, R., & Tobiska, L. (2007). Linear and nonlinear approach to the Rosensweig instability. *GAMM-Mitteilungen*, *30*(1), 171-184.
21. Gollwitzer, C., Rehberg, I., & Richter, R. (2006). Via hexagons to squares in ferrofluids: experiments on hysteretic surface transformations under variation of the normal magnetic field. *Journal of Physics: Condensed Matter*, *18*(38), S2643.
22. Joo, S. W. (2007). A note on the similarity between the normal-field instability in ferrofluids and the thermocapillary instability. *Journal of Fluid Mechanics*, *583*, 459-464.
23. Gomba, J. M., Diez, J., Gratton, R., González, A. G., & Kondic, L. (2007). Stability study of a constant-volume thin film flow. *Physical Review E*, *76*(4), 046308.
24. Moulton, D. E., & Pelesko, J. A. (2010). Reverse draining of a magnetic soap film. *Physical Review E*, *81*(4), 046320.
25. Conroy, D. T., & Matar, O. K. (2015). Thin viscous ferrofluid film in a magnetic field. *Physics of Fluids*, *27*(9), 092102.
26. Conroy, D., & Matar, O. K. (2017). Dynamics and stability of three-dimensional ferrofluid films in a magnetic field. *Journal of Engineering Mathematics*, *107*(1), 253-268.

27. Seric, I., Afkhami, S., & Kondic, L. (2014). Interfacial instability of thin ferrofluid films under a magnetic field. *Journal of Fluid Mechanics*, *755*.
28. Richter, R. (2011). Mag (net) ic Liquid Mountains. *Europhysics news*, *42*(3), 17-19.
29. Finlayson, B. A. (1970). Convective instability of ferromagnetic fluids. *Journal of Fluid Mechanics*, *40*(4), 753-767.
30. Schwab, L., Hildebrandt, U., & Stierstadt, K. (1983). Magnetic bénard convection. *Journal of Magnetism and Magnetic Materials*, *39*(1-2), 113-114.
31. Krakov, M. S., & Nikiforov, I. V. (2002). To the influence of uniform magnetic field on thermomagnetic convection in square cavity. *Journal of Magnetism and Magnetic Materials*, *252*, 209-211.
32. Wen, C. Y., & Su, W. P. (2005). Natural convection of magnetic fluid in a rectangular Hele-Shaw cell. *Journal of magnetism and magnetic materials*, *289*, 299-302.
33. Lange, A. (2002). Thermal convection of magnetic fluids in a cylindrical geometry. *Journal of magnetism and magnetic materials*, *252*, 194-196.
34. Krakov, M. S., Nikiforov, I. V., & Reks, A. G. (2005). Influence of the uniform magnetic field on natural convection in cubic enclosure: experiment and numerical simulation. *Journal of magnetism and magnetic materials*, *289*, 272-274.
35. Suslov, S. A. (2008). Thermomagnetic convection in a vertical layer of ferromagnetic fluid. *Physics of Fluids*, *20*(8), 084101.
36. Nanjundappa, C.E., Shivakumara, I.S.: Effect of velocity and temperature boundary conditions on convective instability in a ferrofluid layer. ASME J. Heat Transf. **130**, 1045021–1045025 (2008).
37. Qin, Y., & Kaloni, P. N. (1994). Nonlinear stability problem of a ferromagnetic fluid with surface tension effect. *European journal of mechanics. B, Fluids*, *13*(3), 305-321.
38. Weilepp, J., & Brand, H. R. (1996). Competition between the Bénard-Marangoni and the Rosensweig instability in magnetic fluids. *Journal de Physique II*, *6*(3), 419-441.
39. Hennenberg, M., Weyssow, B., Slavtchev, S., & Legros, J. C. (2001). Coupling between Marangoni and Rosensweig instabilities. *The European Physical Journal-Applied Physics*, *16*(3), 217-229.
40. Scherer, C., & Figueiredo Neto, A. M. (2005). Ferrofluids: properties and applications. *Brazilian Journal of Physics*, *35*(3A), 718-727.
41. Bitar, A., Kaewsaneha, C., Eissa, M. M., Jamshaid, T., Tangboriboonrat, P., Polpanich, D., & Elaissari, A. (2014). Ferrofluids: from preparation to biomedical applications. *Journal of Colloid Science and Biotechnology*, *3*(1), 3-18.
42. Selyshchev, P. A., Petrenko, V. I., Rajnak, M., Dolnik, B., Kurimsky, J., Kopcansky, P., ... & Bulavin, L. A. (2019). Non-uniform distribution of ferrofluids spherical particles under external electric field: Theoretical description. *Journal of Molecular Liquids*, *278*, 491-495.
43. Nagornyi, A., Petrenko, V. I., Rajnak, M., Gapon, I. V., Avdeev, M. V., Dolnik, B., ... & Timko, M. (2019). Particle assembling induced by non-homogeneous magnetic field at transformer oil-based ferrofluid/silicon crystal interface by neutron reflectometry. *Applied Surface Science*, *473*, 912-917.
44. Kúdelčík, J., Bury, P., Drga, J., Kopčanský, P., Závišová, V., & Timko, M. (2013). Structure of transformer oil-based magnetic fluids studied using acoustic spectroscopy. *Journal of magnetism and magnetic materials*, *326*, 75-80.
45. Selyshchev, P. A., Petrenko, V. I., Rajnak, M., Dolnik, B., Kurimsky, J., Kopcansky, P., ... & Bulavin, L. A. (2019). Non-uniform distribution of ferrofluids spherical particles under external electric field: Theoretical description. *Journal of Molecular Liquids*, *278*, 491-495.
46. SchaÈffer, E., Thurn-Albrecht, T., Russell, T. P., & Steiner, U. (2000). Electrically induced structure formation and pattern transfer. *Nature*, *403*(6772), 874.

47. Schäffer, E., Thurn-Albrecht, T., Russell, T. P., & Steiner, U. (2001). Electrohydrodynamic instabilities in polymer films. *EPL (Europhysics Letters)*, *53*(4), 518.
48. Bae, J., Glogowski, E., Gupta, S., Chen, W., Emrick, T., & Russell, T. P. (2008). Effect of nanoparticles on the electrohydrodynamic instabilities of polymer/nanoparticle thin films. *Macromolecules*, *41*(7), 2722-2726.
49. Shivakumara, I. S., Rudraiah, N., & Nanjundappa, C. E. (2002). Effect of non-uniform basic temperature gradient on Rayleigh–Benard–Marangoni convection in ferrofluids. *Journal of magnetism and magnetic materials*, *248*(3), 379-395.
50. Shivakumara, I. S., & Nanjundappa, C. E. (2006). Marangoni ferroconvection with different initial temperature gradients. *Journal of Energy Heat and Mass Transfer*, *28*(1), 45.
51. Nanjundappa, C. E., Shivakumara, I. S., & Arunkumar, R. (2010). Bénard–Marangoni ferroconvection with magnetic field dependent viscosity. *Journal of Magnetism and Magnetic Materials*, *322*(15), 2256-2263.
52. Nanjundappa, C. E., Shivakumara, I. S., & Arunkumar, R. (2011). Onset of Bénard-Marangoni ferroconvection with internal heat generation. *Microgravity Science and Technology*, *23*(1), 29-39.
53. Nanjundappa, C. E., Shivakumara, I. S., & Srikumar, K. (2013). On the penetrative Benard–Marangoni convection in a ferromagnetic fluid layer. *Aerospace Science and Technology*, *27*(1), 57-66.
54. Kargupta, K., Sharma, A., & Khanna, R. (2004). Instability, dynamics, and morphology of thin slipping films. *Langmuir*, *20*(1), 244-253
55. Kotni, T. R., Sarkar, J., & Khanna, R. (2014). Kinetically engendered subspinodal length scales in spontaneous dewetting of thin liquid films. *Physical Review E*, *90*(2), 020401.
56. Barnes, K. A., Karim, A., Douglas, J. F., Nakatani, A. I., Gruell, H., & Amis, E. J. (2000). Suppression of dewetting in nanoparticle-filled polymer films. *Macromolecules*, *33*(11), 4177-4185.
57. Luo,H.,&Gersappe,D. (2004). Dewetting dynamics of nanofilled polymer thin films. *Macromolecules*, *37*(15), 5792-5799.
58. Besancon, B. M., & Green, P. F. (2005). Polystyrene-based single-walled carbon nanotube nanocomposite thin films: dynamics of structural instabilities. *Macromolecules*, *38*(1), 110-115.
59. Koo, J., Shin, K., Seo, Y. S., Koga, T., Park, S., Satija, S., ... & Rafailovich, M. H. (2007). Stabilizing thin film polymer bilayers against dewetting using multiwalled carbon nanotubes. *Macromolecules*, *40*(26), 9510-9516
60. Xu, L., Yu, X., Shi, T., & An, L. (2009). Investigation of the dewetting inhibition mechanism of thin polymer films. *Soft Matter*, *5*(10), 2109-2116.
61. Mukherjee, R., Das, S., Das, A., Sharma, S. K., Raychaudhuri, A. K., & Sharma, A. (2010). Stability and dewetting of metal nanoparticle filled thin polymer films: control of instability length scale and dynamics. *ACS nano*, *4*(7), 3709-3724.
62. Roy, S., Bandyopadhyay, D., Karim, A., & Mukherjee, R. (2015). Interplay of substrate surface energy and nanoparticle concentration in suppressing polymer thin film dewetting. *Macromolecules*, *48*(2), 373-382.

5. List of Projects submitted/implemented by the Investigators

5.1 Details of Projects submitted to various funding agencies:

S.No.	Title	Cost in Lakh	Month of submission	Role as PI/Co-PI	Agency	Status

1	Study of enhancement of functional properties of PDMS/PS immiscible blends in presence of carbon nanoparticles	Rs.32,55, 840	May, 2019	PI	CSIR	Submitted
2	Bacteria self-assembly driven 3D CNT/ Graphene sponge fabrication for CO2 capture	**Rs.** 16944480 .00	Dec, 2019	**PI**	GAIL	Submitted
3	Efficient Dew harvesting with self-assembled Janus nano-particles embedded over stress-engineered surface	Rs. 67.056 lakhs	Dec, 2019	**PI**	Nanomi ssion	Submitted

5.2 Details of Projects completed during the last 5 years:

Sl. No.	Project Title	Project Duration		Funding Agency	Total Cost
		From	To		
1	Self Organization of complex fluids: Instabilities at soft interfaces; Wet granulates under shear.	2008	2011	Planning Unit, IIT Delhi	Rs. 7 Lakhs
2	Instabilities on Patterned Substrates,Blistering, Fibrillation,Necking and Friction atSoft Thin elastic Films	2010	2013	Department of Science &Technology(DST)	Rs. 29.16 Lakhs
3	CFD Applications in Bioprocessing	2016	2016	Ansys, India	Rs. 10.00 Lakhs
4	Instabilities in wetting thin films with thickness dependent viscosity effects	2014	2018	Department of Science & Technology (DST)	Rs. 30.10 Lakhs

5	Dewetting assisted selfassembly/ origami formation and folding of graphene particles	2018	2018	DST and IUSSTF	Rs.10.11L Akhs

5.3 Details of Projects under implementation:

S. No.	Title	Cost in Lakh	Start Date	End Date	Roles as PI/Co-PI	Agency
1.	Investigation of the interface between electrochemically-active microorganisms and self-assembled graphene/carbon nanotubes over polymeric surfaces.	Rs.10.00Lakhs	2019	2021	PI	IITD

6. List of facilities being extended by parent institution(s)for the project implementation.

6.1 Infrastructural Facilities

Sr. No.	Infrastructural Facility	Yes/No/ Not required Full or sharing basis
1.	Workshop Facility	Not required
2.	Water & Electricity	No
3.	Laboratory Space/ Furniture	No
4.	Power Generator	No
5.	AC Room or AC	No
6.	Telecommunication including e-mail & fax	No
7.	Transportation	Not required
8.	Administrative/ Secretarial support	No
9.	Information facilities like Internet/Library	No
10.	Computational facilities	No
11.	Animal/Glass House	Not required
12.	Any other special facility being provided	Not required

6.2 Equipment available with the Institute/group/ Dapartment/Other Institutes for the project:

Equipment available with	Generic Name of Equipment	Model, Make & year of purchase	Remarks including accessories available and current usage of equipment
PI & his group	Spin coater, Vacuum oven, Microscope, Incubator, Autoclave		All equipment are accessible and in the working condition
PI's Department (Institute)	Rheometer, BET, XRD, optical profilometer		All equipment are accessible and in the working condition
Other Institute(s) in the region	AFM, SEM, TEM, EDX, Raman Spectroscopy, elipsometer		All equipment are accessible and in the working condition

7. Name and address of experts/ institution interested in the subject / outcome ofthe project.

People working in the area of smart materials, polymers and sensors.

(a) **Prof. Ashutosh Sharma** Institute Chair Professor & C .V. Seshadri Chair Professor Secretary, Dept. of Science & Technology, New Delhi **ashutos@iitk.ac.in** **+91-512-259-7026**	(b) **Prof. Rajesh Khanna** Professor Department of Chemical Engineering Indian Institute of Technology Delhi Hauz Khas New Delhi 110 016 **rajkh@chemical.iitd.ac.in** **Tel: (91)-11-2659-1031**
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8. PROPOSED FINANCIAL DETAILS:

1. Proposal Estimated Cost (Rs in Lakh):3854880 (in words Rs. Thirty eight lakh fifty four thousand eight hundred and eighty only)

2. **Budget for Manpower**

Designation	No of Persons	Year1(Rs in Lakh):	Year2(Rs in Lakh):	Year3(Rs in Lakh):	Year4(Rs in Lakh):	Year5(Rs in Lakh):	Total (Rs in Lakh):
Total	1	520800	520800	520800			**15, 62, 400**

3. **Justification for Manpower (200 words)**

The project proposed requires a full time research scholar to complete the objectives of the project. Therefore it is proposed to support **one research scholar** with very good academic background for performing research and development in this area as the project covers one distinct topic. The salary is Rs. 35,000/ with 24% HRA; so the total is 35,000+24%= Rs. 43, 400 X 12 months= Rs. 5, 20, 800 for one person. Rs. 5, 20, 800 X 3 years= Rs. 15, 62, 400

4. **Budget for Consumables**

Items	Year1(Rs in Lakh):	Year2(Rs in Lakh):	Year3(Rs in Lakh):	Year4(Rs in Lakh):	Year5(Rs in Lakh):	Total(Rs in Lakh):
Total	2	2	2			**6**

5**. Justification for Consumables (200 words)**

The project proposed needs to perform some experiments to confirm the theoretical predictions which involve fabrication of nanoparticles and thin film with nanoparticles encapsulated over them. So certain amount of nanoparticles, reagents, substrates, solvents, and polymer are required for synthesis and successful completion of project.

6. **Budget for Travel**

Travel Type	Year1(Rs in Lakh):	Year2(Rs in Lakh):	Year3(Rs in Lakh):	Year4(Rs in Lakh):	Year5(Rs in Lakh):	Total (Rs in Lakh):
Total	1	1	1			**3**

7. **Justification for Travel (200 words)**

The travelling grant will help to attend conferences in the national and international arena to understand the development in the proposed area. It will also help to present the progress in the proposed research to the national and international audience. Workshops related to the research area will help us to be skilled in dealing with sophisticated software and instrument and also help in understanding key findings in the research area.

8. **Budget for Contingency**

Travel Type	Year1(Rs in Lakh):	Year2(Rs in Lakh):	Year3(Rs in Lakh):	Year4(Rs in Lakh):	Year5(Rs in Lakh):	Total: (Rs in Lakh)
Total	1	1	1			**3**

9. **Justification for Contingency (200 words)**

The contingency budget is required for the sudden or unexpected breakdown of system or other cost like AMC, stationaries, characterization of samples and others for the execution of the project.

10. **Budget for Equipment**

Items	Year1(Rs in Lakh):	Year2(Rs in Lakh):	Year3(Rs in Lakh):	Year4(Rs in Lakh):	Year5(Rs in Lakh):	Total(Rs in Lakh):
Total	5	0	0			**5**

11. **Justification for Equipment (200 words)**

The proposed project requires high end computers to simulate the models and achieve results and hence a new computer is to be purchased. The Intel-Fortran compiler is required to compile the results.

12. **Budget for Equipment**

Generic name of Equipment	Imported/Indigenous	Cost in Rs (Lakhs)	Cost in US $ (Lakhs)	Year of Procurement (1/2/3/4/5)
Computer		1		1
Intel-Fortran compiler		4		1
Total		5		

13. Budget for Institute Overhead

Item	Year1(Rs in Lakh):	Year2(Rs in Lakh):	Year3(Rs in Lakh):	Year4(Rs in Lakh):	Year5(Rs in Lakh):	Total: (Rs in Lakh)
Total	284160	184160	184160			**652480**

14. **Justification for Institute Overhead (200 words)**

The institute charges 20% overhead on sponsored projects.

PROFORMA FOR BIO-DATA (to be uploaded)

1. Name and full correspondence address: **JAYATI SARKAR**
2. Email(s) and contact number(s) **40 NORTH AVENUE, INDIAN INSTITUTE OF TECHNOLOGY, DELHI, NEW DELHI:110016**
   Phone: **9717701531**
   Email: **_jayati@chemical.iitd.ac.in**
   WebPage: **http://web.iitd.ac.in/~jayati/**
3. Institution: **DEPARTMENT OF CHEMICAL ENGINEERING, INDIAN INSTITUTE OF TECHNOLOGY, DELHI, NEW DELHI:110016**
4. Date of Birth: **9TH JUNE, 1977**
5. Gender (M/F/T): **FEMALE**
6. Category Gen/SC/ST/OBC: **GENERAL**
7. Whether differently abled (Yes/No): **NO**
8. Academic Qualification (Undergraduate Onwards)

Course	*Year of passing*	*Institution*	*Board/ University*	*Division/ Class*	*CPI/% of Marks*	*Position*
Doctor of Philosophy in Chemical Engineering	2005	Indian Institute of Technology, Kanpur, India	Indian Institute of Technology, Kanpur, India	N/A	CPI **10/10**	N/A
Masters of Technology in Chemical Engineering	2001	Indian Institute of Technology, Kanpur, India	Indian Institute of Technology, Kanpur, India	N/A	CPI **9.67/10**	N/A
B.Che. (Chemical Engineering)	2000	Jadavpur University	Jadavpur University	1st Class (Hons)	**88.55%** (Chemical Engineering as Major)	**1st**

9. Ph.D thesis title, Guide's Name, Institute/Organization/University, Year of Award.

2001-2005 **Indian Institute of Technology,** Kanpur, India

Degree: **Doctor of Philosophy in Chemical Engineering**
PhD Dissertation: *Instabilities of Soft Confined Interfaces: Adhesion,Debonding, Dewetting and Pattern Formation*
Advisor: Professor Ashutosh Sharma
Year of Award: 2006

10. **Work experience** (in chronological order).

S.No	Positions held	Name of the Institute	From	To	Pay Scale
1.	Associate Professor	Indian Institute of Technology, Delhi	2016	2018	148100
2	Assistant Professor	Indian Institute of Technology, Delhi	2008	2016	43250-9000-52250
3.	Postdoctoral Scientist	Max Planck Institute for Dynamics and Self Organization Gottingen Germany	Aug 2007	Aug 2008	
4.	Lead CFD Scientist Chemical	Zeus Numerix Pvt. Ltd,	Feb 2007	Jul 2007	
5.	Senior Research Engineer	Transoft International, Fluidyn Software & Consultancy Pvt. Ltd., Bangalore, India.	Sept 2005	Dec 2006.	

Teaching experience

Duration	Organization	Area(s)
22nd September, 2008onwards	Indian Institute of Technology, Delhi	Chemical Engineering UG and PG Level
2000-2005	Teaching Assistant Dept. of Chemical Engineering, IIT Kanpur, India.	Courses: Fundamentals of Colloid and Interfacial Science and Technology; Introduction to nano-materials and nano-technology; Fluid Mechanics and Rate Processes; and Thermodynamics Responsibility: Conducting exams and grading exam papers.

11. Professional Recognition/ Award/ Prize/ Certificate, Fellowship received by the applicant.

S. NO.	AWARDED FOR	NAME OF AWARD	AWARDING AGENCY	YEAR
1	**To undertake international collaborative research in premier institutions in U.S.A**	**Indo-US WISTEMM fellowship**	**DST**	**2018**
2	**For outstanding contribution**	**Bharat Vikas Award**	**Indian Institute of Self Reliance**	**2017**
3	**Biographical profile for inclusion in the new Who's Who in the World(R) 2016**	**Marquis Who's Who(R)**	**http://www.marquiswhoswho.com**	**2015-2016**

4	**Venus International Research Award**	**Outstanding Scientist Award**	**Venus International Foundation**	**2015**
5	**Best PhD thesis to Student Hemalatha Annepu in the area of Colloid and Interface Sciences under my guidance**	**Shah-Schulman Award**	**IIChE**	**2014**
6	**Excellence in research and development given to scientists under the age of 35 Years**	**Amar Dye-Chem award,**	**IIChE**	**2012**
7	**Contribution in the area of Engineering**	**Kusuma Trust sponsored outstanding "Young Faculty Fellowship" in the area of Engineering at IIT Delhi**	**IIT Delhi**	**2008-2013**
8	Medals Received for standing first in B.ChE, Jadavpur University	• University Medal; • Chemical Engineering Diamond Jubilee Gold Medal; • Chitta Ranjan Khastagir Memorial Gold Medal; • Bandana Ghosh Memorial • Gold Centred Silver Medal; • Jatindra Krishna Memorial Bronze Medal.	Jadavpur University, Kolkata	2000
9	Medals received for Chemical Engineering Viva Examination	Hiralal Roy Smriti Padak (Silver Medal)	Jadavpur University, Kolkata	2000
10	Award received for Merit and Academic Activities	R. Debnath Memorial Award (1999-2000)	Jadavpur University, Kolkata	1999-2000
11	Award received for Merit and Academic Activities	Sumona Memorial Book Prize	Jadavpur University, Kolkata	2000
12	For achieving highest marks Chemistry in Higher Secondary Exam	South Point Prize	South Point High School, Kolkata	1996
13	Good result in Higher Secondary Exam.	National Scholarship	West Bengal council of Higher Secondary Education	1996-1997
14	Good result in Madhyamik Exam.	National Scholarship	West Bengal Board of Secondary Education	1994-1995
*Re*cipient of several other prizes at school level for general proficiency.				
Award received by student: **Hemalatha Annepu: Won the 2nd prize in the department in Open House Poster Presentation for her Phd work, 2013.**				

12. Publications *(List of papers published in SCI Journals, in year wise descending order).*

Citation indices	All
Citations	479
h-index	12
i10-index	12

Publications in refereed journals (international)

Sl. No.	*Title of Paper(s)*	*Name of the Co-Author(s)*	*Name of the Journal*	*Page(s)*	*Volume & Year*
1	Selective adsorption of oil on self-organized surface patterns formed over soft thin PDMS films	S. Basu, J Sarkar	Chemical Engineering Science	970-979	207, 2019
2	Miniaturization of surface patterns in soft elastic film over patterned substrates	S. Basu, J Sarkar	Chemical Engineering Science	195-203	197, 2019
3	Process development in the Quality by Design paradigm: Modeling of Protein chromatography resin fouling	Lalita Kanwar Shekhawat, Mili Pathak, JayatiSakar, Anurag S.Rathore	Journal of Chromatography A,	56-66	1570, 2018.
4	Application of CFD in Bioprocessing: Separation of mammalian cells using disc stack centrifuge during production of biotherapeutics	Lalita K Shekhawat; Jayati Sarkar; Rachit Gupta; Sandeep Hadpe, Anurag Singh Rathore	Journal of Biotechnology	1-11	267, 2018
5	Hierarchical micro-and nanofabrication by pattern-directed contact instabilities of thin viscoelastic films	A Ghosh, D Bandyopadhyay, J Sarkar, A Sharma	Physical Review Fluids	124004	2017
6	A Finite Element Study of Adhesion of Soft Thin Elastic Films cast on Rough Surfaces	S. Mishra,H.Annepu and J. Sarkar	International Journal of Adhesion and Adhesives	102–110	79, 2017
7	Kinetics of sub-spinodal dewetting of thin films of thickness dependent viscosity	T. Rao. Kotni, R. Khanna nad Jayati Sarkar	Journal of Physics: Condensed Matter	175001	29, 2017
8	Failure Regimes of single wet granular aggregate under Shear.	D. Dubey	Journal of Non-Newtonian Fluid Mechanics,	236–248	234, 2016
9	CFD of mixing of multi-phase flow in a bioreactor	Shekhawat, L. K., Loomba, V. and Rathore, A. S.	Biotechnol Progress. doi:10.1002/btpr.2242	613-628	32, 2016

	using population balance model.				
10	**Mechanical Strain Induced Tunable Anisotropic Wetting on Buckled PDMS Silver Nanorods Arrays.**	**P. Goel , S. Kumar , J. Sarkar , and J. P. Singh**	**ACS APPL. MATER. INTERFACES,**	**8419–8426**	**7 (16), 2015**
11	**Miniaturized Pattern Formation in Elastic Films Cast on Sinusoidally Patterned Substrates.**	**H. Annepu**	**LANGMUIR**	**12278-12286**	**30 (41), 2014**
12	**Kinetically engendered subspinodal length scales in spontaneous dewetting of thin liquid films**	**T. Rao Kotni, and R. Khanna.**	***PHYSICAL REVIEW E, RAPID COMMUNICATIONS,* Impact Factor: *2.326* Number of Citations:2 The standards for Rapid Communication are parallel those for PRL but is of interest to a more specialized readership.**	**020401 (R)**	**90, 2014**
13	**Pattern Formation in Soft Elastic Films Cast on Periodically Corrugated Surfacesa Linear Stability and Finite Element Analysis**	**H. Annepu , and S. Basu.**	***Modelling Simul. Mater. Sci. Eng.***	**055003**	***22(5), 2014***
14	**Squeezing instabilities and delamination in elastic bilayers: A linear stability analysis.**	**H. Annepu**	***PHYSICAL REVIEW E***	**051604**	**86, *2012***
15	**Contact Instability of a Soft Elastic Film Bonded to a Patterned Substrate**	**H. Annepu and A. Sharma**	**JOURNAL OF ADHESION**	**214**	**87, 2011**
16	**A Unified Theory of Instabilities in Viscoelastic Thin Films: FromWetting to Confined Films, From Viscous to Elastic Films, From Short To Long Waves.**	**A. Sharma**	**LANGMUIR**	**8464**	**26, 2010**
17	**Electric-field induced instabilities and morphological phase transitions in soft elastic films**	**A. Sharma and V. Shenoy**	**PHYSICAL REVIEW E**	**031604**	**77, 2008**
18	**Electric Field Induced Morphological Transitions in Elastic-Contact-Instability of Soft-Solid Films**	**N. Arun, A.Sharma, V.B.Shenoy and K.S. Narayan**	**JOURNAL OF ADHESION**	**513**	**83, 2007**
19	**Contact Instability in Adhesion and Debonding of Thin Elastic Films.**	**M. Gonguntla, A. Sharma, S.A. Subramanian, M. Ghosh, V.Shenoy**	**PHYSICAL REVIEW LETTERS,**	**018303**	**97, 2006**
20	**Confinement-induced instability and adhesive failure between dissimilar thin elastic films**	**J. Y. Chung, K. Kim, M. K. Chaudhury,**	**THE EUROPEAN PHYSICAL JOURNAL E,**	**47-53**	**20, 2006.**

		and A. Sharma			
22	Adhesion and debonding of s oft elastic films on rough an d patterned surfaces	A. Sharma and V. Shenoy	JOURNAL OF ADHESION	1-25,	81, 2005
22	Adhesion and Debonding of Soft Elastic Films: Crack Pa tterns, Metastable Pathways and Forces.	A. Sharma and V. Shenoy	LANGMUIR,	1457-1469	21, 2005
23	Patterns, forces, and metasta ble pathways in debonding o f elastic films	V. Shenoy and A. Sharma	PHYSICAL REVIEW L ETTERS, Also a selected paper in VIRTUAL JO URNAL OF NANOSCA LE SCIENCE AND NA NOTECHNOLOGY	018302,	93 (1), 200 4. 10 (2), 200 4.
24	Spontaneous surface roughenin ng induced by surface intera ctions between two compressi ble elastic films	V. Shenoy and A. Sharma	PHYSICAL REVIEW E	031607	2003

Manuscripts under review:

1. Self-assembly of graphene nano-particles assisted by dewetting of thin polymer/ polystyrene film by Surita Basu, Bhukya Vishnu Naik, Kalluri Ankarao, Prabir Patra, Jayati Sarkar submitted.

Manuscripts to be submitted shortly (by this semester):

1. The Dynamics of wet disperse phase granulates under shear flow by Dheerendra Dubey, Jayati Sarkar
2. Study of The Dynamics, fluid behavioral index and morphological changes of the polydispsere system by Dheerendra Dubey, Jayati Sarkar
3. Bimodal distribution of thin liquid film morphology for films on coated substrates whose viscosity decreases with film thickness by TirumalaRao Kotni, Jayati Sarkar and Rajesh Khanna
4. Kinetics of thin film dewetting for films whose viscosity inversely varies with film thickness by TirumalaRao Kotni, Jayati Sarkar and Rajesh Khanna.
5. Instabilities in Elastic films having thickness dependant shear modulus, Jayati Sarkar, Sunita Singh
6. Influence of patterned substrates on miniaturization of surface patterns in soft elastic films by Basu, Surita;, Sarkar, Jayati

12A. Details of papers in refereed conferences held abroad:

1. J. Sarkar, Influence of Patterned Substrates on Miniaturization of Surface Patterns in Soft Elastic Films, NANOSMAT 2018, USA.
2. J. Sarkar Miniaturized Pattern Formation at Soft Interfaces, CCMR2017, South Korea.
3. L. Shekhawat, J. Sarkar, A.Rathore, Application of CFD: A case study for bio-separation of microbial cells using disc stack centrifuge in production of biotherapeutics, ACS Conference, USA 2017.
4. S. Mishra, J. Sarkar, Surface roughness effect on Adhesion-Debonding of soft thin elastic film, New Orleans, Louisiana, APS March Meeting, 2017
5. H. Annepu and J. Sarkar; "Ordered miniature pattern formation through elastic contact instabilities", Jülich Soft Matter Days 2015, 10-13 November 2015, Germany.
6 T. Rao Kotni, J. Sarkar, R. Khanna, "Instability and morphology of thickness dependent viscosity liquid films", Interdisciplinary Surface Science Conference, Brimingham,UK, April, 2015.
7. H. Annepu and J. Sarkar; Mesoscale Pattern Formation in Soft Polymers, Adhesion, Science of (GRS), Gordon Research, Conference, USA, 2013.
8. J. Sarkar, "Formation of miniaturized patterns at soft interfaces", 68th IUVSTA Workshop, Hong Kong, 2012.
9. J. Sarkar, Hemalatha Annepu, A. Sharma, Non-
Linear Study of Instabilities of films on Patterned Substrates, International Symposium on Non-

Equilibrium Soft Matter, Nara, Japan, 2010.
10. Kai Huang, Martin Brinkmann, Jayati Sarkar, and Stephan Herminghaus, “Two dimensional wet granul ar matter under shear”, Fachverband Dynamik und Statistische Physik (DY), 2009.
11. J. Sarkar, A. Sharma, and V. Shenoy ; Instabilities at Soft Confined Interfaces, Dynasoft08, Cargese, France, August, 2008.
12. J. Sarkar, Kai Huang, Martin Brinkmann, Jürgen Vollmer, Stephan Herminghaus, "Wet Granulates un der Shear", Dynasoft08, Cargese, France, August, 2008.
13.J. Y. Chung, K. H. Kim, M. K. Chaudhury, J. Sarkar and A. Sharma; "Pattern Formation in Thin Fil ms under Confinement", Adhesion Society Meeting, Jacksonville, Florida, February, 2006.
14 . A. Sharma, J. Sarkar and V. Shenoy; Patterns, forces and metastable pathways in debonding of elasti c films, 27th Adhesion Society meeting, Wilmington, North Carolina from February15-18, 2004.

12 B. Details of papers in refereed conferences held in India

1. Surita Basu, Bhukya Vishnu Naik,Kalluri Ankarao, Prabir Patra, Jayati Sarkar, Graphene nanoparticles in thin films, ICSM, Jaipur, 2018
2. Sinta Singh, Jayati Sarkar, Miniaturizatioj in thin filmes with graded shear modulus, ICSM, Jaipur, 2018.
3. Manga Ramya Durga, Jayati Sarkar, Miniaturization of Patterns under a top attractive surface in case of inhomogeneous thin films, Compflu, IIT Roorkie, 2018
4. J. Sarkar Miniaturized Pattern Formation at Soft Interfaces, INDOGFOE2017, Jaipur.
5.J. Sarkar and S. Sen, Surface instability and morphology in inhomogeneous incompressible elastic thin film, Soft Matter - Young Investigators Meet, SMYIM held at Goa December, 2016.
6. S. Sen and J. Sarkar, "Surface instability and morphology of incompressible elastic thin film having inhomogeneous material properties, Seventh National Level Annual Research Symposium of Chemical Engineering Research Scholar, Chemference, IITGandhinagar, 3-4 Decemder,2016.
7. S. Basu and J. Sarkar, Miniaturization of length scale over patterned substrate, Seventh National Level Annual Research Symposium of Chemical Engineering Research Scholar, Chemference, IITGandhinagar, 3-4 Decemder,2016.
8. S. Basu and J. Sarkar, Experimental observation of minuscule length scale of patterns formed over patterned substrate, ComFlu, IIIT Hyderabad, December 12-14, 2016
9. D. Dubey and J. Sarkar ."Failure regimes of a single wet granular aggregate under shear", Comp Flu, IISER, PUNE, 2016
10. T. Rao Kotni, J. Sarkar, R. Khanna, "Satellite hole formation in thin liquid films exhibiting thickness dependant viscosity: Soft Matter - Young Investigators Meet, SMYIM held at Pondicherry December, 2015
11. D. Dubey and J. Sarkar ."Single Cluster Aggregate under shear flow in Couette ", IUTAM, Symposium in Multiphase Flows with Phase Change Challenge and Opportunities, Hyderabad, December, 2014.
12. J. Sarkar and D. Dubey, Study of the Dynamics of Wet Granulates under Shear, International Confere nce of Powder, Granule and Bulk Solids: Innovations and Applications, November, 2013.
13. D. Dubey and J. Sarkar ."Study the Dynamics and Stress Strain Behaviors of Wet Granulates under Shear", CHEMCON, Jalandhar, Punjab, December, 2012.
14. H. Annepu, J. Sarkar," Miniaturized Pattern Formation at Soft Interfaces", CHEMCON, Jalandhar, Pu njab, December, 2012.
15. H. Annepu, J. Sarkar, “Miniaturization of Patterns in Elastic Films by Substrate Patterning and Bila yers Route”, International Conference on Nanoscience and Technology ICONSAT 2012, Hyderabad, India, 2012.
16 H. Annepu, J. Sarkar, “Varicose Instabilities in Elastic Bilayers”, SERC school and symposium on R heology of Complex Fluids, IIT Guwahati, India, 2012.
17. T.R. Kotni, J. Sarkar and R. Khanna, ‘Quantifying of slippage in thin liquid films’, National conferen ce on chemical engineering, ChemFerence12, 2012.
18. H. Annepu, J. Sarkar, Linear Analysis of Instabilities of films on Patterned Substrates, Chemcon, An namalai University, TamilNadu, India, 2010.
19. J. Sarkar, H. Annepu, A. Sharma, Study of Instabilities of films on Patterned Substrates, Internationa l Conference of Fundamental And Applications of Nano Science and Technology, Jadavpur University, Ko lkata, India , 2010.
20. J. Sarkar, A. Sharma and V. Shenoy. "Stability, Morphology and Forces of Debonding of a soft thin elastic film", National Workshop-cum-Symposium on Complex Fluids , Bangalore, December 3-7, 2002.

12 C Invited talks

Invited talks given by me on my research area of interest:

1. Cluster formation, breakage and rheology of wet granulates under shear- A LBM study, CompFlu-Decemeber-2017, IIT Madras.

2. Award Presentation in CHEMCON, IIT Guwahati, December, 2015.

3. Engineering Challenges in the 21st century a Special Address at Annual Research Meet, VIFARM Chennai, December, 2015

4. Soft Matter - Young Investigators Meet, SMYIM - 2014, being held at Pondicherry. (Also chaired a session in the meeting).

5. Award Presentation in CHEMCON, Jalandhar, Punjab, December, 2012

6. A meet on Industrial Problems Related to Surface & Interfacial Science, Chemical Engineering Depart ment, Indian Institute of Technology Delhi. April 2012.

7. Chemical Industry - Academia Meet, foundation For Innovation and Technology Transfer. IIT Delhi, March 2012.

8. Golden Jubilee Symposium on Fabrication At Small Scales (FASS) & Indo-US Conference on Fabrionics: Science of advanced fabrication, IIT Kanpur, December, 2009

9. Scientific Advisory Board Meeting, Max Planck Institute for Dynamics and Self Organization, 28th Feb ruary, 2008.

10. Max Planck Institute for Dynamics and Self Organization, 30th November, 2007

11. Tata Research Development and Design Centre, Pune, 27th November, 2006.

12. Annual Conference of Chemineer's Society, IIT Kanpur 6th April, 2003.

Invited talks in conferences where my work was portrayed:

1. A. Sharma, J. Sarkar and V. Shenoy; Adhesion, "dewetting" and debonding of soft elastic films: Patter ns, forces and metastability, Workshop on Pattern formation through instabilities in thin liquid films: from fundamental aspects to applications, Max-Planck Institute for the Physics of Complex Systems, Dresden (Germany), September 21-28, 2004.

2. A. Sharma, J. Sarkar and V. Shenoy; Adhesion and debonding of elastic films: patterns, forces and me tastability, STATPHYS-22 Satellite: Pattern Formation in Nonequilibrium Systems, Kolkata, July 11-13, 2004.

3. A. Sharma, J. Sarkar and V. Shenoy; Debonding of a thin soft elastic film, Nanobali 2: Nanotribology: The Liquid-Solid Interface, Bali, 2003.

4. A. Sharma, J. Sarkar and V. Shenoy; Adhesion and debonding of elastic films: Patterns, forces and me tastability, India- Japan Workshop on Surfaces and Interfaces, Saha Institute, Kolkata, 2003.

13. Detail of patents. N/A

14. Books/Reports/Chapters/General articles etc.

Type	Description	Category	Publisher name	Pub. year	Author status
Book chapter	**Simulating contact instability in soft thin Films through Finite Element Technique**	**Authored**	**InTech**	**2016**	**1**

15. Any other Information (maximum 500 words)

Memberships of various governmental, professional or industrial committees including editori al boards of journal, technical conference committees, etc.

1. Editorial Board Member of Scientific Reports

2. Life member of IIChE

3. Reviewer in Journal of Colloids and Interfacial Science, Indian Journal of Chemical Technology, Journal of Chemical Technology & Biotechnology, Mechanics Research Communications, International Journal of Science,

Engineering and Innovative Research etc.
4. Technical Review Committee Member of International Conference of Powder, Granule and Bulk Solids, Patiala, 2013
5. Was appointed by Director, IIT Roorkie as an external eminent expert in V.N.M.M.3 and was asked to evaluate a particular nomination, 2012
6. Was appointed by FITT to evaluate 2 patents done by IIT faculties, 2012.
7. Was appointed as referee to DST projects
8. Was appointed as external examiners of PhD students by Senate Chair of IIT

I have acquired my Bachelor of Chemical Engineering degree from Jadavpur University, Kolkata in the year 2000, M.Tech in Chemical Engineering from IIT Kanpur in the year 2001 and PhD in Chemical Engineering from IIT Kanpur in the year 2005. My PhD research was about study of instabilities in thin soft elastic films and the phenomena that occur there namely adhesion, debonding, dewetting and pattern formation. Pattern formation at soft interfaces is a field of extreme importance and relevance in today's world in fabricating nano-meso scale materials. Apart from Chemical Engineering have also done several advanced level courses in Mechanical and Aerospace Engineering. My subjects of interest are: ***Interfacial Science, Computational Fluid Dynamics, Nanotechnology, Complex Fluids and Solid Mechanics.***

After PhD I have worked for about two years in an MNC and an incubation centre of IIT Mumbai as lead CFD scientist and has worked in areas like dispersed phase flow, multiphase flow, electro kinetics and parallelization. I have completed my postdoctoral studies in the area of Lattice Boltzmann Modeling in complex fluid and granular materials at the Max Planck Institute for Dynamics and Self Organization, Goettingen, Germany in the year 2008, just before joining Chemical Engineering Department in IIT Delhi as Assistant Professor. I am now an Associate Professor in the Department.

In IIT Delhi I have taken up both UG courses like Heat Transfer, Transport Phenomena, Introduction to Chemical Engineering, Fluid Mechanics and Heat Transfer Laboratory, Chemical Engineering Laboratory – I, Design and Laboratory Practices and PG courses like Fundamentals of Computational Fluid Dynamics and Advanced Computational Techniques in Chemical Engineering. In total I have/had 7 PhD students, 6 MTech, 12 Dual Degree and 15 BTech students under my supervision. I have taken up two project from DST and an in-house funded project on complex fluids. I have been a part of Training and placement committee in the department, member of student review committee for PhD students, member of Departmental UG committee, member of Departmental Review committee and dual degree coordinator.

I have been rewarded with several awards of national and international importance for my research output and scholastic achievements. They include University Medals for standing first at BE level in Chemical Engineering, Kusuma Trust sponsored outstanding Young Faculty Fellowship in the area of Engineering at IIT Delhi, (2008), Amar Dye-Chem award, IIChE , 2012 for excellence in research and development given to scientists under the age of 35 Yea rs, Outstanding scientist Award by Venus International Foundation 2015, Marquis Who's Who in the World 2016, Bharat Vikas Award 2017 for outstanding contribution in the field of chosen research and lately have also been awarded by Indo-U.S. WISTEMM-Women Overseas Fellowship 2018. I had been appointed as technical committee member for conferences, reviewer for papers, as an external examiner for several IITK/IITG MS/PhD students. I have lately been given the honor to serve as a member of the Editorial Board for Scientific Reports, a journal from Nature publication.

My thesis has been highly acclaimed by reviewers and the first PhD thesis guided by me has received **Shah-Schulman Award in the field of Colloids and Interfacial Science from IIChE in the year 2014.**